

Spell-Check Engine — Fixes & Improvements

Claims document validation · SPELL check · before & after the rework

ProxyURLApp · commits 22afe5e (9 Jul 2026) & 1e73135 (14 Jul 2026) · report generated 16 Jul 2026

5/20 → 17/20

Forged docs caught on the 20-document reviewer sample

~35 → ~3.5

Flagged tokens per document (noise/false positives)

0

Deploys needed to tune the term list (admin CRUD)

Before: full-dictionary ratio check

The original engine lowercased every 3+ letter token, checked each against the full English dictionary (`nspell` + `dictionary-en`), and marked the claim **FAILED** when more than 20% of words were unknown (`SPELL_MAX_RATIO = 0.2`).

Why it failed in practice — on scanned Indian paperwork the miss-ratio measured OCR quality, not forgery:

- Personal names, place names, and OCR garbage swamped the real typos (~35 flagged tokens per document).
- Lowercasing broke proper-noun lookup — the dictionary knows `India` but not `india` , so every name was "misspelled".
- Acronyms and codes (PAN, HDFC, chassis/VIN prefixes) were all counted as errors.
- Result: only **5 of 20** forged documents in the reviewer sample were caught, buried under false positives.

Fix round 1 — curated expected-term matching commit 22afe5e

Replaced the dictionary-miss ratio with the signal human reviewers actually use: **misspellings of words the form is expected to contain** ("profession" printed as "profesion").

1. **Curated** `EXPECTED_TERMS` **vocabulary** — salary-slip, identity, and insurance/dealer terms. A token that isn't a real word but lands 1–2 edits from an expected term is flagged; names and OCR garbage aren't near any term, so they no longer trigger.
2. **Smarter tokenization** (`spellCandidates`) — case preserved so proper nouns resolve correctly; all-caps acronyms/codes skipped.
3. **Real-word guard** — a token the dictionary accepts (in its own case or lowercased) is never flagged, so "cleaner" near "clerk" is fine.
4. **First-letter-match rule** — real typos alter interior letters ("profeSion"); OCR drops leading characters ("nsurance", "ddress"). Requiring a matching first letter filters that OCR noise.
5. **Capped edit distance** — near-miss check runs in $O(\text{cap} \cdot \text{len})$ not $O(\text{len}^2)$; short terms (<6 chars) match at distance 1 only, since distance 2 on a 4–5 letter word is mostly coincidence.
6. **Actionable findings** — messages changed from `Suspect word: "X"` to `Misspelled "X" (expected "Y")`, each anchored to its OCR word box for in-document highlighting.
7. **Reviewer UX** — misspelled-word summary on hover in the Claims Dashboard; `revalidateClaims` and `setupSpellingChecks` scripts to re-run and demo the check.

Fail threshold at this stage: ≥ 3 distinct expected-term misspellings.

Fix round 2 — accuracy tuning & admin-managed terms commit 1e73135

1. **Threshold lowered 3 → 1** — reviewers flag a form on a single genuine typo ("enginear", "Cleark", "ninty"). Threshold 3 caught 5/20 on the reviewer sample; threshold 1 catches 14/20. Deliberate recall-over-precision trade: SPELL is a pre-filter for human review.
2. **Levenshtein → Damerau (OSA) distance** — an adjacent swap ("nmae" → "name") now counts as one edit instead of two; transpositions are among the most common human misspellings.
3. **Admin-managed term list** — new `SpellTerm` table with CRUD screens in the admin dashboard, seeded from the built-in list. The engine prefers the table and falls back to the built-in list if it's empty or unreadable — the vocabulary is now tunable without a deploy.
4. **Expanded vocabulary** — payroll (conveyance, gratuity, payslip...), education/organisation terms for ID cards, vehicle/finance terms, month names, common reviewer-flagged words, and an entity gazetteer (Bajaj, Jindal, Prudential, Lucknow, Flipkart). This lifted the sample result from 14/20 to **17/20**; the remaining misses are OCR-ceiling cases where Tesseract couldn't read the misspelt word at all.
5. **British-spelling rule** — only en-GB variants are listed ("authorised", never also "authorized"): listing both would flag every correct British spelling as a typo of the American one.
6. **Guardrails** — terms under 4 characters ignored (too noisy); duplicate tokens deduplicated so a header word repeated on every page counts once.

Before / after at a glance

	Before	After
Approach	Every word vs. full English dictionary; fail on >20% unknown	Near-misses of a curated, admin-managed expected-term list; fail on 1 distinct hit
Detection (20-doc sample)	5 / 20 forged docs caught	17 / 20 (rest limited by OCR quality)
Noise	~35 flagged tokens per document	~3.5 per document
Names / acronyms	All flagged (lowercased lookup, no acronym filter)	Ignored (case-aware lookup, all-caps skip, real-word guard)
Distance metric	n/a (exact dictionary membership)	Capped Damerau — transposition = 1 edit
Findings	"Suspect word: X"	"Misspelled X (expected Y)" + page/bbox highlight anchor
Tuning	Code change + deploy	SpellTerm CRUD in admin dashboard, no deploy

Deployment & follow-ups

- Production needs `prisma db push` (new `SpellTerm` table) and the seed run to load the initial term list.
- If legitimate scans produce too many false positives at threshold 1, the documented escape hatches are raising the threshold or gating on OCR word confidence (`WordBox.conf`).
- Extend the term list via the admin screen as new document types or reviewer-cited misses surface — never add both en-GB and en-US spellings of the same word.